

CO4_AT1_DATA INTERPRETATION

1. Real-Time Detection System

A surveillance system uses the following techniques:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Compare the three techniques and evaluate which method should be selected for a surveillance system operating under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

2. Robust Object Detection

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of the three methods under normal, occluded, and low-light conditions. Evaluate which method provides the best overall performance and discuss the trade-offs involved.

3. Optical Flow Analysis

Two optical flow methods are available:

- Method A: Produces stable motion vectors but misses small motions.
- Method B: Detects fine motion but produces noisy vectors.

Compare the two methods and evaluate which method is more suitable for motion tracking in dynamic scenes. Justify your answer based on accuracy, stability, noise sensitivity, and application requirements.

4. Model Generalization and Overfitting – 10 Marks

Model	Training Accuracy	Testing Accuracy
A	98%	70%

B	90%	88%

Compare the two models and evaluate which model is more reliable for real-world deployment. Explain the generalization ability and overfitting risk of each model and justify your selection.

5. System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare the three models and evaluate which model provides the best balance between accuracy, latency, and cost for a real-time application. Justify your choice considering practical deployment constraints.

1. Real-Time Detection System

A surveillance system needs an object detection method that can identify objects accurately while also responding quickly. The three available methods are **Viola-Jones**, **YOLO**, and **Deep Learning**. Their performance is given below:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate

Method	Accuracy	Speed
Deep Learning	95%	Slow

Viola-Jones

Viola-Jones provides **78% accuracy** and has a **fast processing speed**. Its main advantage is that it can provide quick detection. This makes it useful when the surveillance system has very strict time limitations.

However, its accuracy is the lowest among the three methods. In a surveillance system, lower accuracy can lead to missed objects or incorrect detections. Therefore, although it is fast, its detection quality may not be sufficient for applications where accurate identification is important.

YOLO

YOLO provides **92% accuracy** with **moderate speed**. It gives a significant improvement in accuracy compared with Viola-Jones. At the same time, it does not have the slow processing speed of the Deep Learning method.

This makes YOLO a balanced option. A surveillance system can use YOLO to obtain reasonably fast responses while maintaining high detection accuracy. This is particularly useful when the system needs to process information continuously.

Deep Learning

Deep Learning provides the highest accuracy of **95%**. This means it performs the best in terms of detection accuracy among the three methods.

However, its main disadvantage is that it is **slow**. In a surveillance system with strict latency requirements, a slow method may result in delayed detection. Therefore, although Deep Learning provides the best accuracy, it may not be the most practical option when immediate response is required.

Comparison

Viola-Jones is the fastest but has the lowest accuracy. Deep Learning has the highest accuracy but is slow. YOLO lies between the two and provides a good combination of accuracy and speed.

The final decision should depend on the deployment conditions. If the system has very limited computing resources and extremely fast response is required, Viola-Jones may be considered. If maximum accuracy is the main requirement and slower processing is acceptable, Deep Learning can be used.

However, if the system needs **both good accuracy and reasonable speed**, YOLO is the better choice.

Conclusion

YOLO should be selected for the surveillance system. It provides 92% accuracy with moderate speed. It gives much better accuracy than Viola-Jones while avoiding the slow processing associated with Deep Learning. Therefore, YOLO provides the best practical balance between **accuracy, speed, and real-time requirements**.

2. Robust Object Detection

Robust object detection means that the detection method should continue to perform well even when the environment becomes difficult. The three methods are tested

under **normal conditions, occlusion, and low-light conditions.**

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Normal Conditions

Under normal conditions, **Deep Learning performs best with 95% accuracy.** YOLO achieves 90%, while Viola-Jones achieves 80%.

This means that Deep Learning can identify objects more accurately when there are no major environmental difficulties. YOLO also performs well and is only slightly lower than Deep Learning.

Occlusion Conditions

Occlusion occurs when an object is partly hidden by another object or obstruction.

Under occlusion, Deep Learning gives **85% accuracy**, YOLO gives **75%**, and Viola-Jones gives only **60%**.

This shows that Deep Learning is better at handling partially hidden objects. Viola-Jones experiences a significant reduction in performance, falling from 80% under normal conditions to 60% under occlusion.

Low-Light Conditions

Low-light conditions make object detection more difficult.

Deep Learning achieves **80% accuracy**, YOLO achieves **70%**, and Viola-Jones achieves only **55%**.

Again, Deep Learning performs best. Viola-Jones shows the largest reduction in accuracy and therefore appears to be less suitable for difficult lighting conditions.

Overall Comparison

Deep Learning has the highest accuracy in all three conditions:

- Normal: **95%**
- Occlusion: **85%**
- Low Light: **80%**
- YOLO gives:
- Normal: **90%**
- Occlusion: **75%**
- Low Light: **70%**
- Viola-Jones gives:
- Normal: **80%**
- Occlusion: **60%**
- Low Light: **55%**

Therefore, Deep Learning is the most robust method based on the given data.

Trade-Offs

The main advantage of Deep Learning is its high accuracy and robustness. It maintains relatively strong performance even under difficult conditions.

However, a highly accurate method may require greater computational resources. Therefore, deployment conditions should also be considered.

YOLO provides a good alternative because its performance is also strong across all conditions. Viola-Jones is simpler and faster but loses considerable accuracy when objects are occluded or when lighting is poor.

Conclusion

Deep Learning provides the best overall performance because it has the highest accuracy under normal, occluded, and low-light conditions. It is therefore the best choice when robustness and accuracy are the main priorities. However, YOLO can be considered when a balance between performance and practical deployment requirements is needed.

3. Optical Flow Analysis

Optical flow is used to analyze the movement of objects between images or video frames. The question provides two methods:

Method A: Produces stable motion vectors but misses small motions.

Method B: Detects fine motion but produces noisy vectors.

Both methods have advantages and disadvantages.

Method A

Method A produces **stable motion vectors**. Stability is important because it allows the system to track an object's movement smoothly.

For example, when an object moves continuously, stable vectors can help the system estimate its direction without unnecessary changes. This reduces the possibility of unstable tracking.

However, Method A has a major limitation: it **misses small movements**. Therefore, if an application requires detection

of fine or subtle movements, Method A may not provide sufficient information.

Method B

Method B is designed to detect **fine movements**. This means it can identify small changes that Method A may fail to detect.

This makes Method B useful for applications where small movement is important. However, its major disadvantage is that it produces **noisy motion vectors**.

Noise can make the motion information unstable. Instead of showing only the actual movement, the system may detect unwanted changes. This can affect the accuracy of tracking and may result in an incorrect estimation of an object's path.

Comparison

Factor	Method A	Method B
Stability	High	Lower
Fine-motion detection	Lower	High

Factor	Method A	Method B
Noise sensitivity	Lower	Higher
Tracking smoothness	Good	May be unstable
Small movement detection	Limited	Good

Method A is better when **stable and smooth tracking** is required.

Method B is better when **detecting small movements** is more important.

Selection for Dynamic Scenes

Dynamic scenes can contain several moving objects and changes in movement. In such situations, stable motion information is important for reliable tracking.

Therefore, if the main objective is to track objects smoothly and consistently, **Method A is more suitable.**

However, if the application specifically needs to detect small movements, Method B may be better. In that situation, additional techniques for reducing noise could be used.

Conclusion

Method A is recommended for general motion tracking in dynamic scenes because of its stable motion vectors and lower noise sensitivity. Method B is useful when fine-motion detection is more important, but its noisy output can make tracking less stable.

Therefore, the final selection should depend on the application requirement: **stability → Method A; fine-motion sensitivity → Method B.**

4. Model Generalization and Overfitting

The two models have the following training and testing accuracies:

Model	Training Accuracy	Testing Accuracy
Model A	98%	70%

Model	Training Accuracy	Testing Accuracy
Model B	90%	88%

This comparison is useful for understanding **generalization and overfitting**.

Model A

Model A has a very high training accuracy of **98%**. This means it performs extremely well on the data used during training.

However, its testing accuracy is only **70%**. The difference between training and testing accuracy is:

$$98\% - 70\% = 28 \text{ percentage points}$$

This is a very large difference.

Such a large gap indicates a strong possibility of **overfitting**. Overfitting occurs when a model learns the training data too specifically and therefore does not perform well when it receives new or unseen data.

Therefore, Model A may appear highly accurate during training, but its performance is much weaker in real-world situations.

Model B

Model B has a training accuracy of **90%** and testing accuracy of **88%**.

The difference is:

$$90\% - 88\% = 2 \text{ percentage points}$$

This is a very small difference.

This indicates that Model B performs consistently on both training and testing data. Therefore, it has better **generalization ability**.

Generalization means that a model can learn useful patterns from training data and apply those patterns successfully to new data.

Comparison

Model A:

- Training accuracy = 98%
- Testing accuracy = 70%
- Difference = 28%
- High overfitting risk
- Poorer generalization

Model B:

- Training accuracy = 90%
- Testing accuracy = 88%
- Difference = 2%
- Low overfitting risk
- Better generalization

Although Model A has a higher training accuracy, this does not mean that it is the better model.

For real-world deployment, the model must perform well on data that it has not seen before. Therefore, testing accuracy and the difference between training and testing performance are very important.

Conclusion

Model B is more reliable for real-world deployment. It has a much smaller difference between training and testing accuracy, indicating better generalization and lower overfitting risk.

Model A should not be selected based only on its 98% training accuracy because its testing accuracy is only 70%. Model B provides more consistent and reliable performance.

5. System Performance under Constraints

The three models have different accuracy, latency, and cost values:

Model	Accuracy	Latency	Cost
Model A	88%	50 ms	Low
Model B	92%	80 ms	Moderate
Model C	95%	120 ms	High

A real-time application must consider more than accuracy. **Latency and cost** are also important.

Model A

Model A provides **88% accuracy** with only **50 ms latency** and **low cost**.

Its major advantage is speed. A latency of 50 ms means that the system can respond quickly. Its low cost also makes it suitable for applications with limited budgets.

However, its accuracy is the lowest of the three models.

Therefore, Model A is suitable when **speed and low cost are more important than maximum accuracy**.

Model B

Model B provides **92% accuracy**, **80 ms latency**, and **moderate cost**.

Its accuracy is higher than Model A by 4 percentage points. At the same time, its latency is still relatively low compared with Model C.

Model B therefore provides a balanced solution. It does not have the lowest latency, but its 80 ms latency may be acceptable for many real-time applications.

Its moderate cost also makes it more practical than Model C.

Model C

Model C provides the highest accuracy of **95%**.

However, it has **120 ms latency** and **high cost**.

The high accuracy is an advantage when detection errors are very important. However, the high latency may be a disadvantage for real-time applications that require quick responses.

The high cost may also make large-scale deployment difficult.

Overall Comparison

Model A is best in terms of:

- Lowest latency
- Lowest cost

- Fast response

Model C is best in terms of:

- Highest accuracy
- Model B provides:
- Good accuracy
- Reasonable latency
- Moderate cost

Therefore, Model B provides the best overall balance.

Practical Deployment

If the application requires the **fastest possible response** and has a limited budget, Model A can be selected.

If the application requires **maximum accuracy** and can accept higher cost and latency, Model C can be selected.

However, for a normal real-time application where all three factors matter, Model B is the most practical option.

Conclusion

Model B is recommended for the real-time application because it provides 92% accuracy, 80 ms latency, and moderate cost. It gives better accuracy than Model A without the high latency and cost of Model C.

Therefore, Model B represents the best compromise between **accuracy, response time, and cost**.